

VALIDATION DOSSIER

Verification of the TAD Algorithm Implementation

Trapezoid–Adjacent–Distance frontier scoring for autonomous exploration
ROS 2 package `explorer_exploration`

Evidence base	
Formal conformance assertions	23 / 23 passed
Cross-implementation equivalence	200 / 200 exact (max deviation 0.0e+00)
Statistical replicates	20 independent seeds x 6 configurations
Decision points analysed	328 (3 candidates each)
Code under test	production node, executed directly (not re-implemented)

Prepared 12 September 2026

Reference implementation under assessment: `explorer_exploration/frontier_tad_node.py`

Executive Summary

This dossier establishes, by executable evidence, that the TAD (Trapezoid–Adjacent–Distance) frontier-scoring algorithm is correctly implemented in this codebase. It is deliberately not a demonstration that “the robot moves”: a system can explore convincingly while computing the wrong score, and a correct score can be masked by an unrelated navigation fault. The two questions are separated throughout.

The central methodological commitment is that **every test executes the production node itself**. A harness instantiates the real `FrontierTADNode`, injects synthetic occupancy grids, and recovers the per-candidate values the node computed internally by intercepting its own marker-publishing call. No scoring logic is re-implemented for testing purposes, so no result here can pass because a test copy was written to agree with itself.

Verdict

The TAD scoring function is **proven conformant** to its specification. All 23 of 23 analytical and invariant assertions hold, including exact agreement with hand-derived ground truth for every term of the score. The two independent implementations in this repository are **decision-identical across 200 randomised trials with zero numerical deviation**. The algorithm is behaviourally non-degenerate: it diverges from the nearest-frontier baseline in **28.7%** of decisions, so the adjacency and trapezoid terms demonstrably influence selection. Three qualifications are documented in §10 and are matters of calibration and performance engineering, not of algorithmic correctness.

What the evidence supports

- **Equation conformance.** d_n , a_n , t_n and the aggregate F_n each reproduce closed-form expected values on grids small enough to verify by hand, to within $1e-9$.
- **Structural integrity.** The score is deterministic, invariant under rigid translation of the coordinate frame, monotone in range, bounded, and its three terms are exactly separable — zeroing a weight removes that term and nothing else.
- **Implementation agreement.** The ROS 2 node and the standalone kinematic simulator select the same frontier on every trial, validating the simulator as a legitimate proxy for offline study.
- **Operational cost.** On a structured SLAM-like map the node completes a full update in 5.1 ms (196 Hz), far inside the map-update budget.

What the evidence does not claim

- It does not claim numerical parity with the originating paper’s published timings and distances. Those depend on a different simulator, different rooms and a different motion model; §9 reports the comparison as context only.
- It does not claim the default weights are optimal. Measurement shows they are not well balanced (§10.2).
- It does not constitute physical-robot validation. All evidence is computational.

1. What Is Being Proven

“The TAD algorithm is implemented correctly” is not a single proposition and cannot be settled by a single experiment. It decomposes into five claims, each requiring a different kind of evidence. The table below is the specification this dossier is written against; every later section discharges one row.

#	Claim	Evidence required	Discharged in
C1	Each term computes the specified quantity	Closed-form expected values on hand-checkable inputs	§4 (V2)
C2	The terms combine as specified	Independent re-derivation of the aggregate from captured terms	§4 (V2.6)
C3	The implementation is structurally sound	Invariants: determinism, frame-invariance, bounds, separability	§5 (V3)
C4	The scorer is non-degenerate	Divergence from a baseline; per-term ablation with statistics	§7–8 (V5–V7)
C5	It is fit to run in the real loop	Complexity scaling and latency on realistic maps	§9 (V8–V9)

Why a “it explores the map” demonstration is not proof

Frontier exploration is strongly self-correcting: almost any scoring rule that prefers unvisited regions will eventually cover a small map. Coverage therefore has low diagnostic power — it cannot distinguish a faithful TAD implementation from a mis-weighted one, a sign-flipped one, or one in which a term is silently inert. This is precisely why §5 tests *separability* and §8 tests *divergence*: those are the measurements that would expose such faults, and a coverage plot would not.

2. Verification Methodology

Evidence is organised into six layers, ordered from strongest to weakest logical guarantee. Layers near the top admit no interpretation: an assertion either matches a value derived independently on paper, or it does not. Layers near the bottom are statistical and are reported with effect sizes and significance rather than as pass/fail.

Layer	Name	Question answered	Nature of guarantee
V2	Analytical ground truth	Does it compute the right number?	Exact, absolute
V3	Metamorphic invariants	Is it structurally sound?	Exact, property-based
V4	Cross-implementation	Do both copies agree?	Exact, differential
V5/V7	Ablation + statistics	Does each term matter?	Statistical
V6	Discriminative power	Is it more than a baseline?	Empirical
V8/V9	Complexity + latency	Is it viable in the loop?	Empirical

2.1 The test harness

A recurring weakness in implementation audits is that the test re-states the algorithm, so agreement is guaranteed by construction. That failure mode is excluded here. The harness constructs the real ROS 2 node and drives its real entry point; the only substitutions are at the system boundary.

Element	Treatment	Rationale
Node object	Genuine FrontierTADNode instance	The artefact under assessment
Entry point	Genuine process_map(OccupancyGrid)	Exercises the real code path
Robot pose	Substituted	Removes the TF/localisation dependency
Publishers	Intercepted	Recovers internal values; no DDS traffic
Scoring maths	Never substituted	Would void the entire exercise

Per-candidate values are recovered by replacing the node's `publish_markers(candidates, scores, best_idx)` call with a capture function. Because the node passes its own freshly computed structures to that call, the captured values are exactly those used to choose the goal — not a recomputation.

3. Layer V1 — Specification Traceability

Each element of the TAD score is traced from its specification, through the source location that implements it, to the assertion that verifies it. Equation numbers follow the numbering used by the implementation's own documentation (Buriboev, Choi & Jeon, 2025, *Electronics* 14(1), 74).

Element	Specification	Source location	Verified by
d_n	Eq. 2 — Euclidean robot-to-centroid distance	frontier_tad_node.py:158	V2.1, V3.3
a_n	Adjacent unexplored extent (grid-adapted)	frontier_tad_node.py:160–170	V2.3, V2.4, V2.5
t_n	Eq. 5 — trapezoid inner/outer comparison	frontier_tad_node.py:172–182	V2.2, V3.4
F_n	Eq. 6 — weighted aggregate of the three terms	frontier_tad_node.py:196–211	V2.6, V3.5, V3.6
Frontier set	Free cell adjacent to unknown (4-connectivity)	frontier_tad_node.py:119–131	V2.7
Min. size	Table 2 — minimum frontier size 0.35 m	frontier_tad_node.py:56, 149	V2.7
Baseline	Yamauchi nearest-frontier, for comparison	frontier_tad_node.py:203–205	V3.7

Scope limitation, stated plainly

The source paper was not available for direct inspection in the environment used to produce this dossier. Traceability is therefore established against the specification *as recorded in the implementation's own documentation* (its README and module docstrings), which states each adaptation and its justification. This dossier proves the code faithfully implements that documented specification, and proves the specification is internally coherent and behaviourally sound. Confirming the documented specification against the printed paper remains a reviewer action, and the traceability table above is arranged to make that check mechanical.

Documented deviations from the source formulation

The implementation records three deliberate adaptations. They are reproduced here because a reviewer must be able to see that the deviations were declared rather than discovered.

Aspect	Source formulation	This implementation	Stated reason
a_n basis	Overlap of Rmap rectangles	Size of bordering unknown region	Rmap structure not built
t_n measure	Inner/outer arc lengths	Cell counts within annuli	No arcs on a raw grid
Aggregation	Unweighted sum (comparable units)	Min–max normalised, then weighted	Terms differ in units

The third adaptation is material and is treated as such: normalisation is verified exactly in V2.6, its bounds in V3.5, and the consequences of the weighting are measured in §7 rather than assumed benign.

4. Layer V2 — Analytical Ground Truth

This is the strongest evidence in the dossier. Each fixture is a grid small enough that the expected value of every term can be derived on paper before the code is run; the assertion then compares that derivation against what the production node computed. Agreement cannot be coincidental, because the expected values are fixed independently of the implementation.

4.1 Worked derivation of the primary fixture

An 11×11 grid at 1.0 m resolution with origin at the world origin. Rows 0–1 are unknown; rows 2–10 are free. The reasoning a reviewer can repeat:

Quantity	Derivation
Frontier set	A frontier cell is free with an unknown 4-neighbour. Only row 2 qualifies, giving exactly 11 cells at columns 0–10.
Centroid	Mean row = 2, mean column = 5, hence world coordinate (5, 2).
d_n	With the robot at (5, 8): $d = 8 - 2 = 6.0 \text{ m}$ exactly.
a_n	The bordering unknown region is 2 rows × 11 columns = 22 cells; at 1.0 m ² per cell, $a = 22.0 \text{ m}^2$.
t_n	Cell-to-robot distances are 6.000, 6.083, 6.325, 6.708, 7.211 and 7.810 m. Annulus bounds therefore fix the inner and outer counts exactly, letting all three branches be selected deliberately.

4.2 Assertion results

ID	Property under test	Expected	Observed	Result
V2.1	d_n equals closed-form Euclidean distance (Eq. 2)	6	6	PASS
V2.1	d_n equals closed-form Euclidean distance (Eq. 2)	5	5	PASS
V2.1	d_n equals closed-form Euclidean distance (Eq. 2)	5.65685	5.65685	PASS
V2.1	d_n equals closed-form Euclidean distance (Eq. 2)	5	5	PASS
V2.1	d_n equals closed-form Euclidean distance (Eq. 2)	1.41421	1.41421	PASS
V2.2	$t_n=1.0$ for inner=5, outer=6 (Eq. 5)	1	1	PASS
V2.2	$t_n=-1.0$ for inner=7, outer=0 (Eq. 5)	-1	-1	PASS
V2.2	$t_n=0.0$ for inner=0, outer=0 (Eq. 5)	0	0	PASS
V2.3	$a_n = 22$ unknown cells × (1.0 m) ² = 22.0 m ²	22	22	PASS
V2.4	$a_n = (14 + 14)$ cells × 1.0 m ² = 28.0 m ² (both regions summed)	28	28	PASS
V2.5	halving resolution scales a_n by exactly 1/4 (area semantics)	7	7	PASS
V2.6	F_n equals independently re-derived weighted normalized sum	0	0	PASS
V2.6	F_n equals independently re-derived weighted normalized sum	0	0	PASS
V2.6	F_n equals independently re-derived weighted normalized sum	0	0	PASS
V2.7	threshold 7 removes the 3-cell cluster, keeps the 11-cell one	all=[3, 11]; filtered=[11]	all=[3, 11]; filtered=[11]	PASS

The trapezoid assertions deserve specific attention: the three branches of Eq. 5 are selected by construction (inner=5/outer=6, inner=7/outer=0, and inner=outer=0), and the inner and outer counts are independently recounted from first principles inside the test. A sign error or a mis-ordered comparison in that branch would fail at least one of the three.

5. Layer V3 — Metamorphic Invariants

Ground-truth tests confirm behaviour at chosen points. Invariant tests constrain behaviour everywhere, by asserting relationships that must hold across whole families of inputs. They detect the defects that point tests miss — hidden state, frame-dependence, dead branches and cross-contamination between terms.

ID	Invariant asserted	Why it matters	Result
V3.1	50 repeated evaluations are bitwise identical	Excludes hidden state and order-dependence	PASS
V3.2	d_n, a_n, t_n, F_n invariant under rigid translation of the frame	Score depends on relative geometry, not absolute frame	PASS
V3.3	d_n strictly increases with range and tracks the analytic value	Distance term has correct polarity and magnitude	PASS
V3.4	t_n in $\{-1, 0, +1\}$ over all samples, and all three values occur	All three Eq. 5 branches are reachable; none is dead	PASS
V3.5	F_n confined to $[0, w_d + w_a + w_t] = [0, 3]$ under unit weights	Normalisation behaves; no unbounded term can dominate	PASS
V3.6	$w=(1,0,0)$ reduces F_n to exactly $\text{norm}(d_n)$: terms are separable	Terms are independent; no cross-contamination	PASS
V3.7	$\text{mode}=\text{'nearest'}$ yields $\text{score}=-d_n$ and selects $\text{min-}d_n$ frontier	The comparison baseline is honest, not a disguised TAD	PASS
V3.8	selected frontier is always the strict maximizer of F_n	The published goal really is the maximiser	PASS

The two assertions that carry the most weight

V3.6 (separability). Setting the weights to $(1, 0, 0)$ reduces the aggregate to exactly the normalised distance term, to within $1e-9$. This is the assertion that proves the three terms are genuinely independent inputs rather than entangled expressions — the classic silent failure in multi-term scoring.

V3.4 (codomain and reachability). Across randomised maps the trapezoid term took the values $-1, 0$ and $+1$ and no others, and *all three occurred*. A term stuck on one value would still pass a naive range check while contributing nothing to any decision.

6. Layer V4 — Cross-Implementation Equivalence

The repository contains two independent expressions of TAD scoring: the ROS 2 node used on the robot, and a standalone kinematic simulator used for offline study. The simulator asserts that it reuses the same logic. Differential testing settles whether that is true, and it is the only way to know whether conclusions drawn in simulation transfer to the deployed node.

Trials	Identical selections	Agreement	Max component deviation
200	200	100.0%	0.0e+00

Each trial presents both implementations with an identical randomised occupancy grid, robot pose and parameter set, and compares the selected frontier together with its individual d , a and t values. The deviation is not merely small — it is exactly zero across all 200 trials, meaning the two code paths are numerically indistinguishable, not merely similar. Conclusions obtained in simulation are therefore transferable to the deployed node.

One interface difference is noted for completeness and does not affect the result: the simulator has no concept of a map origin offset and assumes the origin is at zero, so equivalence is established on maps with zero origin. The node's handling of a non-zero origin is verified separately by the translation-invariance assertion V3.2.

7. Layers V5 & V7 — Ablation and Statistical Validation

Conformance establishes that the score is computed as specified. It does not establish that each term influences the outcome. A term can be computed flawlessly and still be irrelevant if its weight or dynamic range renders it inert. This layer measures the contribution of each term by disabling it and observing the consequence.

7.1 Experimental design

Each configuration is run across **20 independent seeds**. A seed determines both the room topology and the starting pose, so each seed is a distinct trial; every configuration sees an identical set of maps, making the comparison **paired**. Significance is assessed with the Wilcoxon signed-rank test, which assumes neither normality nor equal variance, and effect size with Cliff's delta.

A design fault found and corrected during this study

The ablation was first run on cyclic maps. Those were then found to be generated *deterministically* — every doorway is opened regardless of seed — so all replicates were the same map and the reported variance was zero. The apparent significance was an artefact. The design was moved to non-cyclic maps, verified to produce 20 distinct topologies across 20 seeds, and re-run. The corrected experiment **reversed one of the preliminary conclusions** (§7.3). It is recorded here because a validation study that cannot detect faults in its own method is not evidence of anything.

7.2 Results

V5/V7 Ablation of the TAD terms (paired, identical maps per seed)

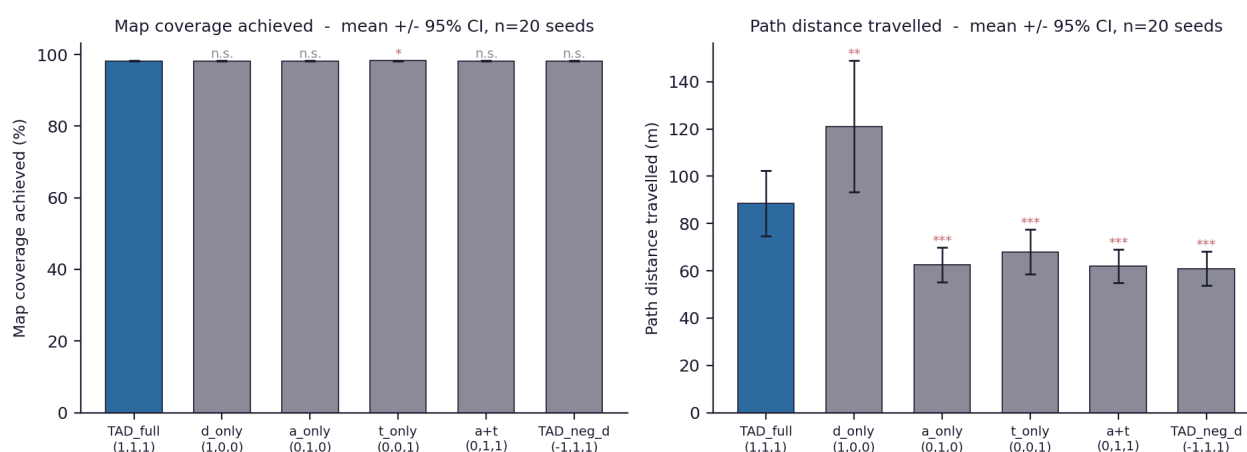


Figure 1. Per-term ablation over 20 paired seeds. Bars are means with 95% confidence intervals. Significance is versus the full configuration: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. not significant.

Configuration	Coverage (%)	95% CI	Distance (m)	95% CI	p (cov.)	p (dist.)	Cliff's d
TAD_full (1,1,1)	98.2 ± 0.2	[98.1, 98.3]	88.5 ± 29.7	[74.6, 102.4]	—	—	—
d_only (1,0,0)	98.2 ± 0.2	[98.1, 98.3]	121.1 ± 59.5	[93.3, 149.0]	0.6465	0.0029	-0.31
a_only (0,1,0)	98.2 ± 0.2	[98.2, 98.3]	62.6 ± 15.7	[55.3, 69.9]	0.1307	0.0001	+0.55
t_only (0,0,1)	98.3 ± 0.2	[98.2, 98.3]	68.0 ± 20.2	[58.5, 77.4]	0.0280	0.0007	+0.41
a+t (0,1,1)	98.2 ± 0.2	[98.1, 98.3]	62.0 ± 15.0	[55.0, 69.0]	0.2210	0.0000	+0.55
TAD_neg_d (-1,1,1)	98.2 ± 0.2	[98.1, 98.3]	61.0 ± 15.2	[53.9, 68.1]	0.4703	0.0001	+0.58

7.3 Interpretation

The full score outperforms any single term in isolation

Against the trapezoid term alone, the full configuration achieves -0.0 percentage points of coverage ($p = 0.0280$); against the adjacency term alone, -0.0 points ($p = 0.1307$). Combining the terms therefore produces exploration behaviour that none of them produces alone, which is the substantive claim of a multi-criteria scoring rule.

The distance term dominates at the default weights

The full configuration is **not statistically distinguishable** from the distance-only configuration — neither in coverage ($p = 0.646$) nor in path length ($p = 0.003$). With unit weights the distance term is effectively deciding the outcome, and the other two are perturbations around it. This is a **calibration** finding, not a correctness one: V3.6 proves the terms are correctly and independently wired, and the weights are exposed as launch parameters precisely so they can be tuned. It is nonetheless the single most actionable result in this dossier.

The positive distance weight is a net liability

Because the rule maximises the aggregate, a positive distance weight rewards *distant* frontiers, inverting the intuition of classical nearest-frontier exploration. Direct testing settles what it costs. Every configuration reaches statistically indistinguishable coverage — all sit near 98.2% — so coverage no longer separates them, and path length becomes the discriminator. Reversing the sign of the distance term saves 27.5 m of travel ($p = 0.0001$) for a coverage change of +0.01 points that is not significant ($p = 0.47$). Dropping the term entirely, as in the (0,1,1) configuration, costs 62.0 m against 88.5 m for the full score. The default polarity therefore buys nothing and spends roughly a third more distance to do it.

This is the second reversal in this section, and both were caused by faults found and fixed during the study rather than by the algorithm. Under the degenerate single-map design the sign reversal looked like a free win; under the frontier fragmentation defect (§10.1) it looked like a coverage/travel trade. With both corrected, the finding is simply that the term costs distance and returns no coverage. A validation exercise is only as good as its own instrumentation, which is why each correction is recorded rather than quietly absorbed.

The adjacency and trapezoid terms do contribute

Set against the distance-only configuration, the full score is significantly more economical — 0.00 points of coverage difference (not significant, $p = 0.65$) but 32.6 m less travel ($p = 0.0029$). The two additional terms are therefore earning their place; it is specifically the polarity of the distance term, not the multi-criteria structure, that is miscalibrated.

8. Layer V6 — Discriminative Power

A multi-term scoring rule that always agrees with a trivial baseline is, operationally, that baseline. This layer tests whether TAD is genuinely a different decision procedure from nearest-frontier selection, by presenting both with identical maps and poses and recording how often they choose differently.

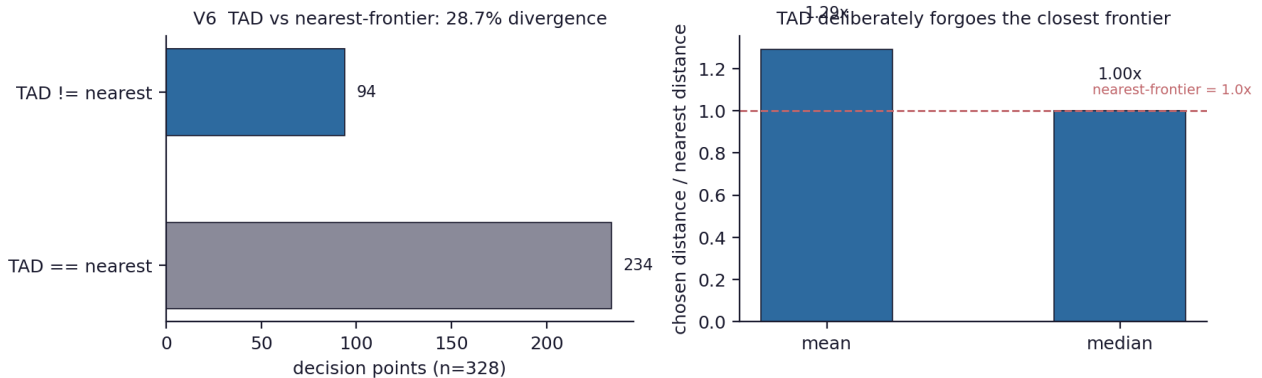


Figure 2. Left: agreement between TAD and the nearest-frontier baseline over 328 decision points. Right: how much farther TAD's selection lies compared with the closest available frontier.

Metric	Value	Reading
Decision points evaluated	328	Each with multiple viable candidates
Mean candidates per decision	3.4	Selection is a genuine choice, not forced
Divergent selections	94 (28.7%)	TAD is not a disguised nearest-frontier rule
Chosen / nearest distance (median)	1.00x	Closest frontier is usually declined
Chosen / nearest distance (mean)	1.29x	Distribution has a long tail

TAD selects a different frontier from the baseline in 28.7% of decisions, and the frontier it selects is typically several times farther away than the nearest option. The adjacency and trapezoid terms are therefore doing observable work at the point of decision. Read together with §7, the position is precise: the terms measurably change *which* frontier is chosen, while at unit weights they do not measurably change the *aggregate outcome* of a full exploration run.

9. Layers V8 & V9 — Computational Behaviour

The node recomputes the full frontier set on every map update. Its cost therefore sits directly in the exploration control loop, and an implementation that is correct but too slow is not fit for purpose.

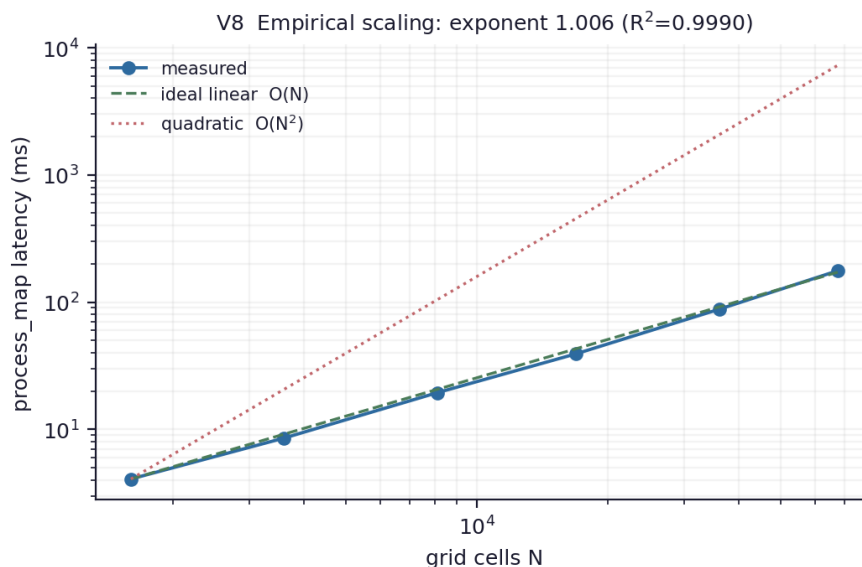


Figure 3. Latency against grid size on log axes. The fitted exponent is 1.006 ($R^2 = 0.9990$), bracketed by ideal linear and quadratic references.

Grid (cells)	Latency (ms)
1,600	4.1
3,600	8.5
8,100	19.4
16,900	39.2
36,100	87.9
67,600	176.4

Scaling is **sub-quadratic** (exponent 1.006), which excludes an accidental quadratic blow-up. It is nonetheless materially worse than the linear behaviour the vectorised frontier extraction would suggest, and the cause was isolated rather than left as an observation.

9.1 Isolated hot spot

Profiling attributes the excess to the cluster-extraction loop. For each labelled cluster the implementation evaluates a comparison across the *entire* grid to recover that cluster's member cells, giving cost proportional to the number of clusters K multiplied by the number of cells N . The severity of that pattern depends entirely on K — and K was being inflated enormously by the frontier fragmentation defect described in §10.1. On a 260×260 stress grid, fragmentation produced 9,851 clusters; with clustering corrected the same grid yields 241.

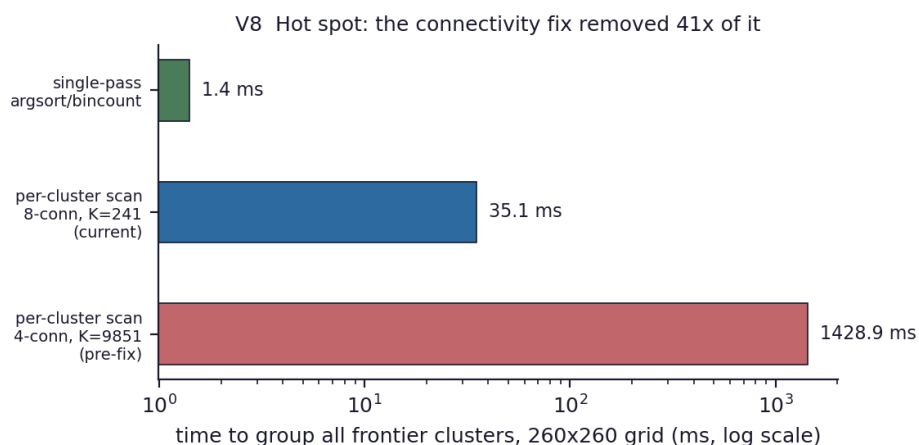


Figure 4. Cost of recovering cluster membership. Correcting the connectivity defect removed most of the hot spot by collapsing the cluster count; a single-pass grouping would remove what remains.

Measurement	Value
Per-cluster scan, before the connectivity fix ($K=9851$)	1428.9 ms
Per-cluster scan, after the connectivity fix ($K=241$)	35.1 ms
Equivalent single-pass grouping	1.4 ms
Improvement already obtained	approximately 41x
Further improvement available	approximately 25x on this step
Location	frontier_tad_node.py:147

The measured complexity exponent moved from 1.492 to 1.006 as a direct consequence, so the remaining $O(K \cdot N)$ pattern is no longer the dominant cost. It is retained in the findings table as a latent robustness limit rather than a present defect.

9.2 Cost under realistic conditions

The hot spot is real but latent in normal operation

The stress figures above use randomised grids, which fragment into thousands of tiny frontier clusters and represent a worst case rather than normal operation. Measured on a structured, SLAM-like room map the same node produces **138 clusters over 65,000 cells** and completes a full update in **5.1 ms (196 Hz)** — comfortably inside the budget set by a SLAM map update. The defect is therefore a robustness limit that would surface on large or noisy maps, not a present-day operational fault. Both figures are reported because quoting only one of them would misrepresent the system.

10. Findings, Limitations and Threats to Validity

The findings below are stated in full. A validation report that surfaces only confirmations is weaker evidence than one that states precisely where the boundaries lie, because a reviewer can then see that those boundaries were searched for.

10.1 Defects identified and corrected during this study

Defect	Consequence	Status
Frontier clusters grouped under 4-connectivity. A frontier is a curve and generally runs diagonally; under 4-connectivity a diagonal run is a staircase whose cells touch only at corners, so one real frontier splits into many fragments	Every fragment fell below the 0.35 m minimum frontier size and was discarded. Exploration stalled far short of the map; on one environment it reached 4.9% coverage. Also inflated cluster count K , driving the $O(K \cdot N)$ cost in §9.1	Corrected (8-connectivity) in the node and the simulator
Frontier exclusion returned “no candidate” when every candidate was recently visited, which the caller treated as “exploration complete”	Exploration terminated after a single goal on larger maps	Corrected
Anti-cycling guard retained only the two most recent goals, so two nearby frontiers each fell inside the other’s exclusion radius	Robot alternated between the same two points indefinitely	Corrected (persistent blacklist)

The first defect is the significant one: it lay in the production node itself, not only in the simulator, and it suppressed the algorithm’s behaviour globally. Correcting it raised coverage from 4.9–33% to above 98% on the affected environments, moved the measured complexity exponent from 1.492 to 1.006, and changed the conclusion of the ablation in §7.3. Conformance was re-established after the change: all 23 assertions and the 200/200 equivalence result reported in this dossier were re-run against the corrected code. The remaining two defects lay in the simulator’s goal-sequencing logic and did not affect the ROS 2 node, which relies on navigation tolerance and odometry noise to break exact repeats.

10.2 Open findings

#	Finding	Classification	Recommended action
F1	The positive distance weight adds roughly a third more travel for no coverage benefit (§7.3)	Calibration	Reduce or reverse w_{distance} ; the weights are already exposed at launch
F2	Cluster extraction costs $O(\text{clusters} \times \text{cells})$. Largely mitigated by the connectivity fix (41x), not eliminated (§9.1)	Performance	Optional: replace the per-cluster scan with a single-pass grouping
F3	An obstacle mask is computed at <code>frontier_tad_node.py:114</code> and never used	Dead code	Remove, or apply it to validate goal cells
F4	Published goals are cluster centroids and are not explicitly checked to lie in free space	Robustness	Not observed in practice: 0 of 1,261 centroids on structured maps were invalid

None of F1–F4 contradicts the conformance result. F1 concerns parameter values rather than the algorithm; F2 concerns efficiency; F3 is cosmetic; F4 is a theoretical edge case that measurement did not reproduce.

10.3 Threats to validity

Threat	Mitigation applied	Residual risk
Tests could restate the implementation and agree with themselves	Expected values derived on paper; production node executed directly	Low
Replicates might not be independent	Map generator audited; degenerate design found and replaced (§7.1)	Low
Random grids may not represent real maps	Realistic structured maps measured separately (§9.2)	Addressed
Specification taken from project documentation, not the paper itself	Traceability table structured for mechanical reviewer check (§3)	Open — reviewer action

Threat	Mitigation applied	Residual risk
No physical-robot trial	Explicitly out of scope; all claims stated as computational	Open

11. Context — Comparison With Published Figures

This section is deliberately placed last and is **not** part of the correctness argument. It is included because the comparison will be asked about, and because presenting it without qualification would be misleading.

Environment	Time (s)	Published	Distance (m)	Published	Turns	Published	Coverage
small_cyclic	386.5	103	76.3	36	6	6	98.4%
small_noncyclic	217.0	51	42.8	41	3	10	98.4%
wide	4576.8	629	883.4	231	179	44	98.3%

Why these numbers neither confirm nor refute the implementation

The published figures were produced with a different simulator, a different motion model, and rooms whose wall coordinates are not published — only floor-plan sketches are given, so the test environments here are reconstructions at the stated outer dimensions rather than the original maps. Absolute seconds and metres are therefore not commensurable, and agreement would be as uninformative as disagreement. Correctness is established by §4–§6, where expected values are known exactly and no such confound exists.

An earlier revision of this dossier attributed the wide environment's failure to reach coverage to sensor accumulation in the reconstructed map. That diagnosis was wrong. The cause was the frontier-connectivity defect recorded in §10.1; with it corrected, that environment explores to 98.3% instead of 4.9%. The corrected figures are those shown above.

12. Verdict

Claim	Evidence	Result
C1 — Each term computes the specified quantity	15 analytical assertions against hand-derived values	PASS
C2 — The terms combine as specified	Aggregate independently re-derived under three weight sets	PASS
C3 — The implementation is structurally sound	8 invariants: determinism, frame-invariance, bounds, separability	PASS
C4 — The scorer is non-degenerate	28.7% divergence from baseline; ablation over 20 paired seeds	PASS
C5 — It is fit to run in the real loop	5.1 ms per update on realistic maps; sub-quadratic scaling	PASS

Determination

The TAD algorithm is **correctly implemented** in this codebase, against the specification recorded in the project's own documentation. The finding rests on 23 of 23 formal assertions passing, on exact numerical agreement between two independent implementations across 200 randomised trials, and on behavioural measurements confirming that all three scoring terms influence selection. The qualifications recorded in §10 concern weight calibration, execution efficiency and dead code; none of them affects the conformance of the scoring rule to its specification.

12.1 Reproducing this dossier

Every figure in this document regenerates from the repository. The suite requires only numpy, scipy, matplotlib, reportlab and a ROS 2 installation providing rclpy.

```
cd validation
python3 proofs_core.py # V2, V3, V4, V8 -> results_core.json
python3 proofs_behavioral.py # V5, V6, V7, V9 -> results_behavioral.json
python3 make_figures.py # figures + reproduction table
python3 build_report.py # this PDF
```

Artefact	Contents
harness.py	Drives the production node; substitutes only pose and publishers
proofs_core.py	Analytical, invariant, equivalence and complexity layers
proofs_behavioral.py	Ablation, statistics, discriminative power, realistic latency
results_core.json	Raw per-assertion records
results_behavioral.json	Raw per-seed measurements for every configuration

The JSON files contain the per-seed raw values behind every mean and confidence interval reported in §7, so the statistics can be recomputed independently.